

Filtering and Subqueries

Start @ 9:03

Recap:

Query types.

Relationships.

F.K. P.K.

Order by

Join table

limit, offset

Order of exec.

functions.

Alias

inline calculation.

Agenda.

Concat

lower, upper.

String func.

where clause

like,

distinct

null

case statement

if, else,

sub-queries.

String fn: Do some operation on string

round(3.48932, 2) → 3.49

concat.

fname	lname

concat(fname, lname)
↓

Therich Batcha

John Henry

Concat(Upper(Fname), " ", lower(Lname))

Substr ("Tharish", 2, 3) →

↑ ↑ ↑

Start no. of
pos char to
 extract

thanish batcha $\rightarrow$ substa (1, 1)
 $\uparrow$ $\uparrow$ $\leftarrow$ substa (2,) lower

`initcap("thanish")` $\longrightarrow$ `Thanish`

Where: Filter rows based on a condition
get all from cat 1

only grows where $\text{cat} = 1$

Cat	Period:
	Bicubit

F	2	Cookies
T	1	Water
T	3	Tea
F		

1	Cookies
1	Water

Select * from DB-table
Where col = "A"

→ select all columns
and only rows
where the col has
the value "A".

Order of writing

Select
from
where
Order by
limit
offset

1000000 ← From
10000 ← where
1000 ← Select

Order of exec.

Order by
offset
limit

Give v-id = 3 and date between 4 and 9.

Vendor id	Date	v-id = 3	date ≥ 4	date ≤ 9	AND	AND
1	1	F	F	T	F	
2	2	F	F	T	F	

3	3	T	F	T	T
3	4	T	T	T	T
3	5	T	T	T	T
4	6	F	T	T	T
5	7	F	T	T	T
6	8	F	T	T	T
5	9	T	T	T	T
3	10	F	T	F	T

$T \text{ AND } T \rightarrow T$
 $T \text{ AND } F \rightarrow F$
 $F \text{ AND } T \rightarrow F$
 $F \text{ AND } F \rightarrow F$

$T \text{ OR } T \rightarrow T$
 $T \text{ OR } F \rightarrow T$
 $F \text{ OR } T \rightarrow T$
 $F \text{ OR } F \rightarrow F$

$\text{NOT } T \rightarrow F$
 $\text{NOT } F \rightarrow T$

Name	= Diaz	= Edwards	= Wilson	And And	OR OR
Diaz	T	F	F	T	T
Edwards	F	T	F	T	T
Wilson	F	F	T	T	T
Armen	F	F	F	T	F

Wildcards: Filter based on partial string.

Username
Jerome
Jeromiah
Jerry
Adam
Tem

Like.

% → 0 or any number of chars, could be any char

_ → only 1 char, but could be any char.

like "jer %" → the first 3 chars has to be "jer" and then there can be any char or need not be.

"jer _" → the first 3 chars has to be "jer", but there has to be only 4 words the 4th char can be anything

"G _ _ d" → Strictly 4 chars.
• Strictly start with "G"
and end with "d"

1st and 3rd can be any char

"a%" → Strictly start with "a".
strictly min 1 char.

max → can be any length

"% or %" → can be any number of chars
but should contain "or"
either beginning or ending
or middle.

ex: author^s, Orange, landlord

Name	Date
ABC	Jan 1
ABC	Jan 2
ABC	Jan 2
XYZ	Jan 2

select distinct name, date

ABC	Jan 1
ABC	Jan 2
XYZ	Jan 2

distinct name	not distinct date
ABC	Jan 1
XYZ	Jan 1
	Jan 2
	Jan 2

select distinct name

ABC
XYZ

Order of writing

Order of execution

Order of clauses

Select
distinct

From

Where

Order by

limit

Offset

From

Where

Select

distinct

Order by

Offset

limit